

Binary Batch Distillation

Final Lab Report
Unit Operations Lab 5
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1. Executive Summary (10 pts)

The main purpose of this experiment was to analyze the separation performance of a binary batch distillation column for a 10 wt% ethanol water mixture and to evaluate how operating conditions influence boil up rate, pressure drop, product composition and column efficiency. The two experiments were performed using the Armfield UOP3CC continuous distillation column. The reboiler power was varied to change the boil-up rate, and the reflux ratio was adjusted to observe its effect on product purity.

Higher reboiler power increased the boil-up rate from 1.88 L/hr at 0.75 kW to 3.75 L/hr at 1.25 kW, which improved vapor-liquid contact and raised the column efficiency from 22.1% to 31.0%. However, the higher power also increased the pressure drop in the column from 75 to 115 cm H₂O, and caused more foaming on the trays, making the system less stable at higher vapor flow. When the reflux ratio was increased from 5:1 to 12:1, the distillate became slightly richer in ethanol 90% to 93% v/v, while the bottom product became more water rich. With lower reflux, product was collected faster, but at a lower purity

Overall, the results show that better separation is achieved with higher boil up rates and higher reflux ratios, but these improvements come with a tradeoff and that tradeoff is having to use higher energy and potential column instability. Effective operation requires balancing product purity with energy use and maintaining stable flow conditions.

2. Introduction (10 pts)

Distillation is one of the most widely used separation processes in chemical engineering and is often the first method considered when a separation problem arises. It can handle a wide range of feed compositions and produce high-purity products across numerous industries, including petrochemical, pharmaceutical, and food processing. Despite the development of alternative techniques, distillation remains the dominant method for liquid separations due to its versatility, efficiency, and adaptability to large-scale operations [2]. In binary systems, such as ethanol and water, column performance is

evaluated using parameters such as boil-up rate, pressure drop, and column efficiency; each of which directly influences product purity and energy usage.

This experiment utilized a batch distillation column to study the effect of reboiler power on the performance of a binary ethanol–water system. Specifically, the objective was to determine how changes in reboiler duty influenced the boil-up rate, vapor–liquid equilibrium (VLE) behavior, and overall column efficiency. The 10 wt% ethanol–water feed was separated using the Armfield UOP3CC Continuous Distillation Column, which contains sieve trays to enhance vapor–liquid contact. Reboiler power was varied between 0.75 kW, 1.0 kW, and 1.25 kW to evaluate how increased heat input affected vapor generation and separation performance. The compositions of the distillate and bottoms products, as well as the corresponding VLE data, were used with the Fenske equation to estimate the number of theoretical plates and column efficiency.

In Experiment B, the same ethanol–water mixture was distilled at a fixed reboiler power while adjusting the reflux ratio. This allowed investigation of how varying the proportion of condensed vapor returned to the column affected distillation efficiency and product composition.

Overall, this study provided a clearer understanding of how both energy input (through reboiler power) and operating conditions (through reflux ratio) influence column behavior. Such insights are vital for designing efficient distillation systems that balance purity, throughput, and energy consumption in industrial operations.

3. Theory (10 pts)

Distillation separates components based on volatility differences. When a liquid mixture is heated, the components with the higher vapor pressure at a given temperature become enriched in the vapor phase. In the ethanol-water system, ethanol is the more volatile component, so the vapor formed during boiling contains a greater mole fraction of ethanol than the liquid. Repeated cycles of vaporization and condensation increase the ethanol concentration in the distillate product [3].

In the distillation column used in this lab, there are sieve trays that create conditions for direct contact between the rising vapor and the descending liquid. It is assumed that for an ideal system, there would be vapor-liquid equilibrium between various phases per tray. In other words, there would be one theoretical stage for each tray[3]. In real distillation columns, perfect equilibrium cannot be attained because there are limits to the rate of mass transfer. In those situations, distillation tray efficiency can be estimated. There are various formulas to define distillation tray efficiency. One method to define distillation tray efficiency uses Murphree tray efficiency:

$$\eta = \frac{y_n - y_{n+1}}{y_n^i - y_{n+1}}$$

Where y_n is the vapor leaving a tray, y_{n+1} is the vapor entering the tray below, and y_n^i is the equilibrium vapor composition corresponding to the liquid phase on that tray [4].

The ease of separating two components is based on the relative volatility which is defined using the equilibrium vapor-liquid ratios $K_i = \frac{y_i}{x_i}$:

$$\alpha_{EtOH/H_2O} = \frac{K_{EtOH}}{K_{H_2O}}$$

Higher relative volatility means fewer stages needed for separation. Under total reflux, all condensed liquid is returned to the column, and no distillate is removed. This condition produces the maximum separation possible for a fixed number of trays and is used to determine the theoretical number of stages. The required number of stages could be found using the Fenske equation [4]:

$$n+1 = \frac{\log \left[\left(\frac{x_{EtOH}}{x_{H_2O}} \right)_d \left(\frac{x_{H_2O}}{x_{EtOH}} \right)_b \right]}{\log(\alpha_{avg})}$$

Once the theoretical stage count is determined, the overall column efficiency is:

$$E = \frac{\text{Theoretical Stages}}{\text{Actual Number of Trays}} * 100\%$$

The McCabe–Thiele method is a graphical approach for determining the number of theoretical stages. Vapor composition (y) is plotted against liquid composition (x) using vapor–liquid equilibrium data. An operating line is drawn from mass balance considerations, and theoretical stages are obtained by stepping between the operating line and the equilibrium curve. In continuous distillation, the feed location creates two operating lines, but in batch distillation, there is no feed, so the entire column behaves as a single section with one operating line [3].

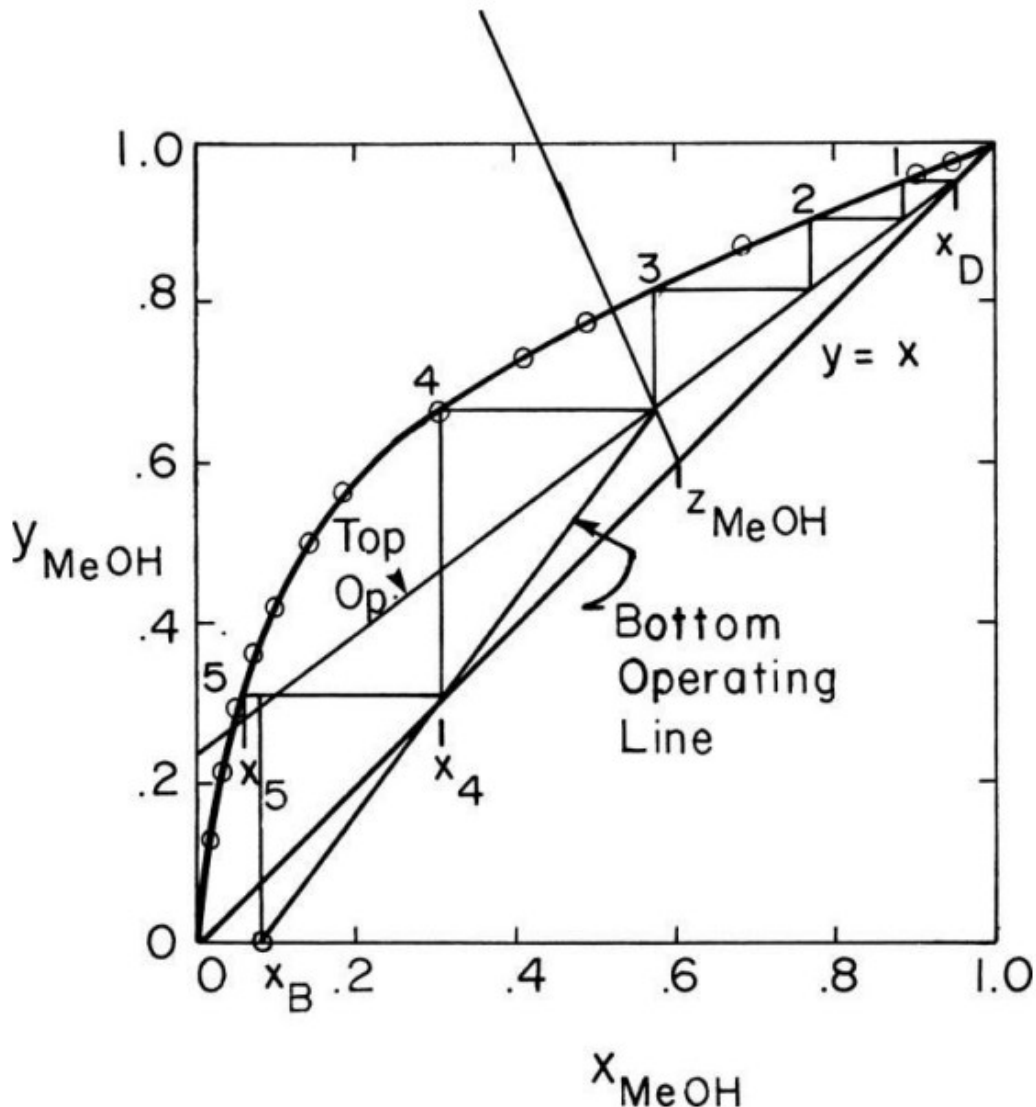


Figure 1: McCabe–Thiele Diagram [5]

Batch distillation is an unsteady-state process. A fixed mixture is charged to the reboiler at the beginning, and no liquid is withdrawn from the bottom during operation. As distillation proceeds, ethanol is preferentially removed, and the reboiler composition

gradually shifts toward higher water content. Consequently, the distillate composition decreases over time, which is observed in Experiment B [1].

The boil-up rate, determined by the reboiler heat input, controls the vapor flow and tray pressure drop. At low boil-up rates, vapor-liquid contact is weak, and separation is limited. As boil-up increases, tray froth height increases, and mass transfer improves. Excessive boil-up can lead to entrainment or flooding, reducing separation performance [6].

The measured distillate and bottom compositions are used to determine the number of theoretical stages via the Fenske equation, and the known number of trays is used to calculate overall column efficiency. The boil-up rate and pressure drop measurements are used to identify operating regions and evaluate how reboiler power influences hydrodynamics and separation behavior.

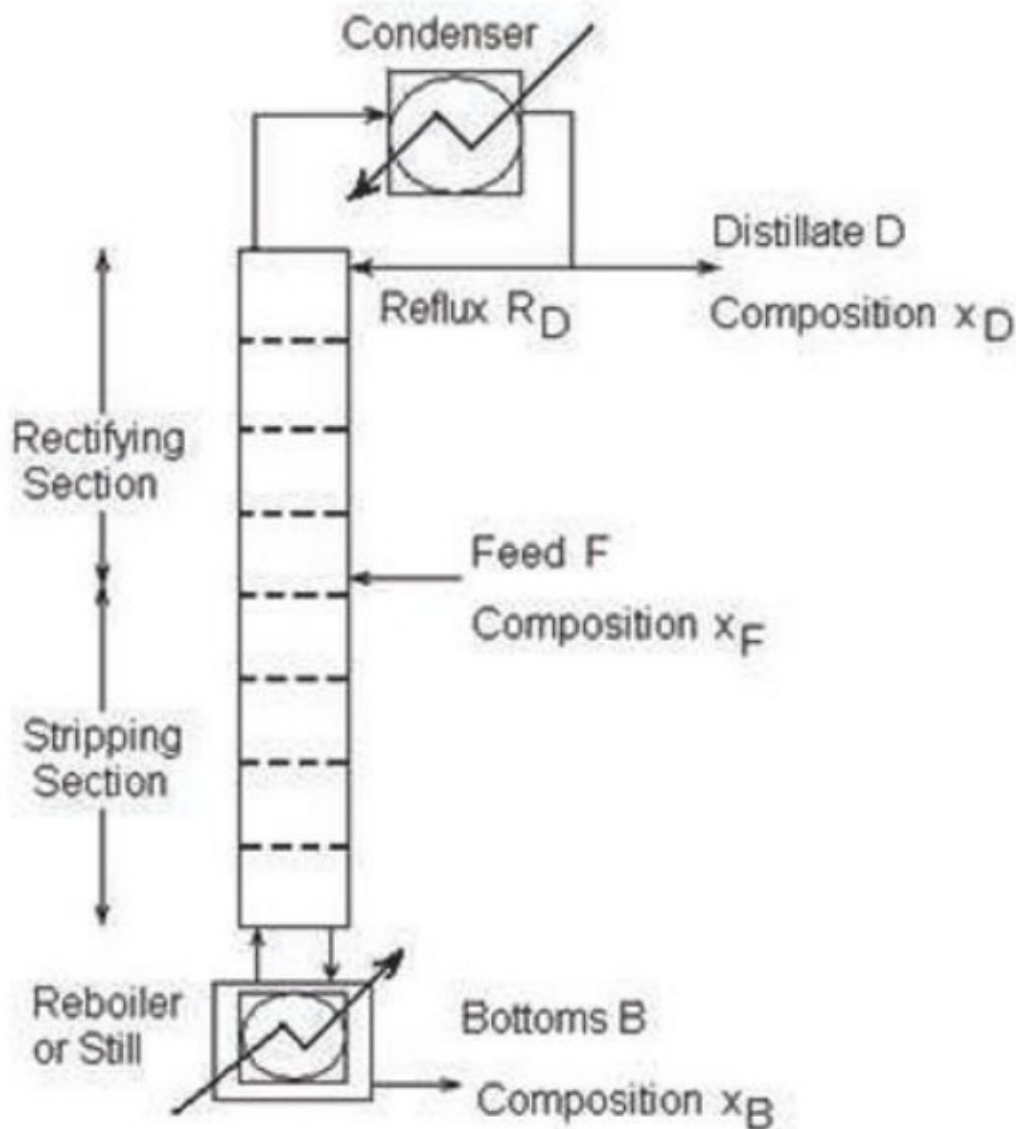


Figure 2: Schematic of Distillation Column [1]

Table 1: Units

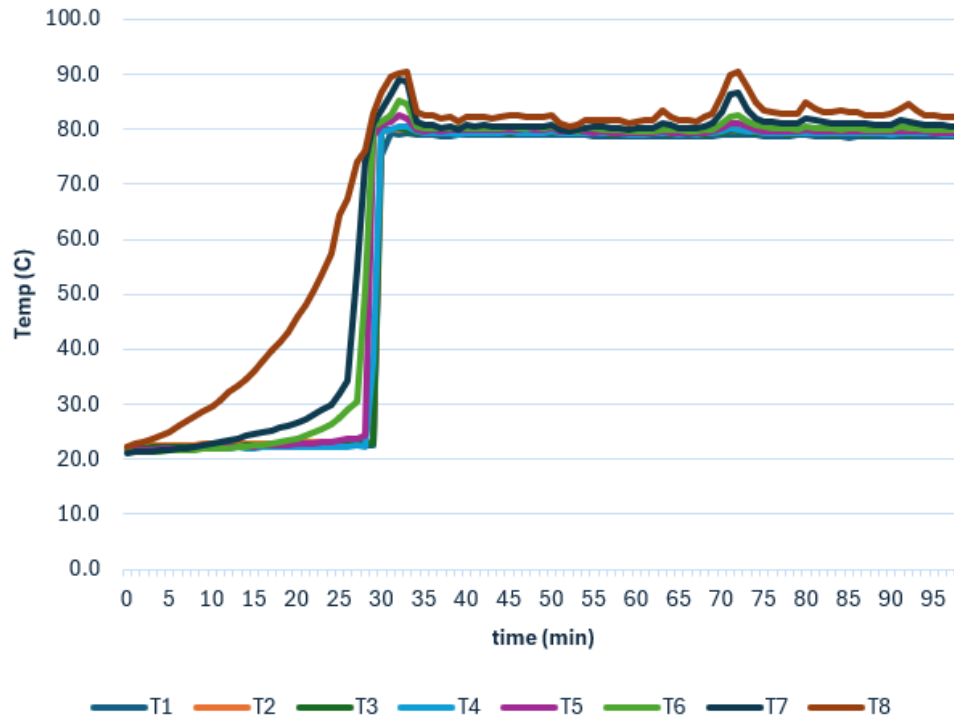
QUANTITY	UNITS
Reboiler Duty (Q)	kW
Boil-Up Rate ($\dot{V}$)	L/hr
Pressure Drop (ΔP)	In H ₂ O
Distillate Composition(xd)	%v
Bottom Composition (xb)	%v
Vapor-Liquid Equilibrium Ratio K_i	Dimensionless
Relative Volatility (α)	Dimensionless
Tray Temperature (T)	°C
Reflux Ratio (R)	Dimensionless
Column Efficiency (E)	%
Number of Stages (n)	count
Time (t)	Min or s

4. Results (15 pts)

Experiment A

This experiment investigated the influence of reboiler power on boil-up rate, pressure drop, and overall column behavior for a binary ethanol–water system operating under total reflux. A heating element within the reboiler supplied adjustable power to generate vapor, while eight thermocouples distributed along the glass column measured tray temperatures, where the bottom tray was measured by T8 and the top tray by T1. The system began at 1.25 kW to accelerate startup and was gradually reduced to 1.0 kW and 0.75 kW after steady state was reached at each setting. The data collected were used to characterize column hydrodynamics and thermal response under varying heat input.

Figure 3 displays the temperature vs. time profile for the entire experiment. The solution reached its boiling point at approximately 30 minutes, as indicated by a sharp temperature rise across all thermocouples. Once boiling began, temperatures quickly stabilized between 80 °C and 83 °C, consistent with the expected boiling range of the ethanol–water mixture near atmospheric pressure. The bottom trays responded first, while upper trays followed with a short lag, showing the upward propagation of vapor and heat through the column. After about five minutes, a uniform steady-state temperature profile was established and maintained for the remainder of the run.

Figure 3: Temperature vs Time for the entire experiment

Figures 4–6 show the temperature profiles for each reboiler power setting. At 1.25 kW, the temperature of the bottom tray (T8) remained the highest due to proximity to the reboiler, while the top tray (T1) was the lowest, reflecting vapor enrichment in ethanol. As the duty decreased to 1.0 kW and 0.75 kW, overall temperatures slightly declined, but the same thermal gradient pattern persisted. Each profile shows minor oscillations in tray temperatures, corresponding to fluctuations in vapor flow and condensate return during steady reflux operation.

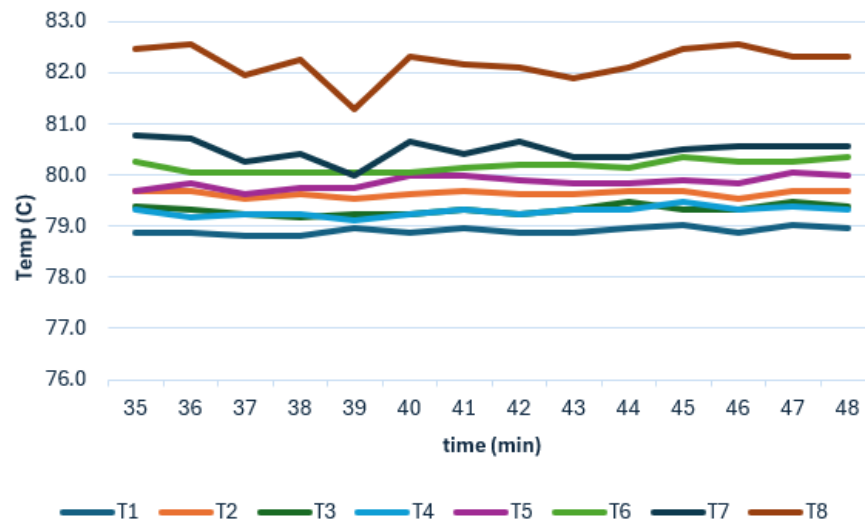
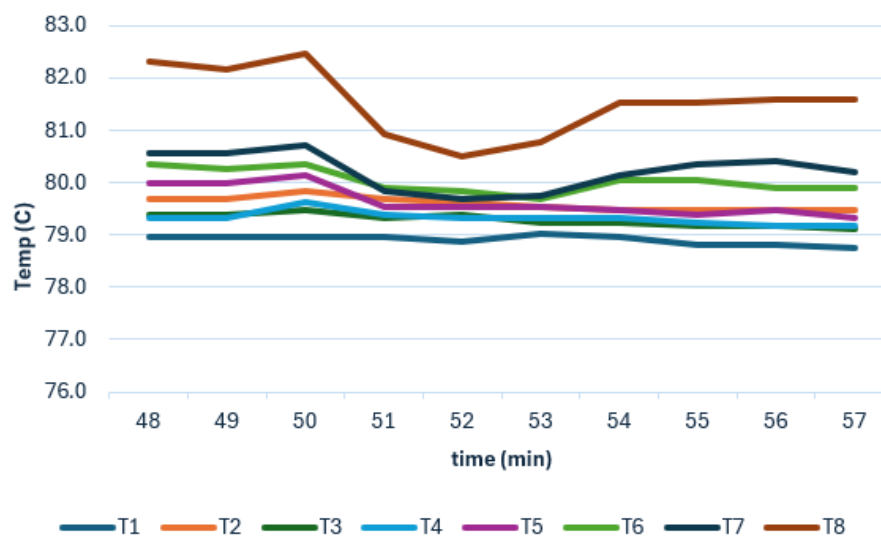
Figure 4: Temperature vs Time for 1.25 kW*Figure 5: Temperature vs Time for 1.0 kW*

Figure 6: Temperature vs Time for 0.75 kW

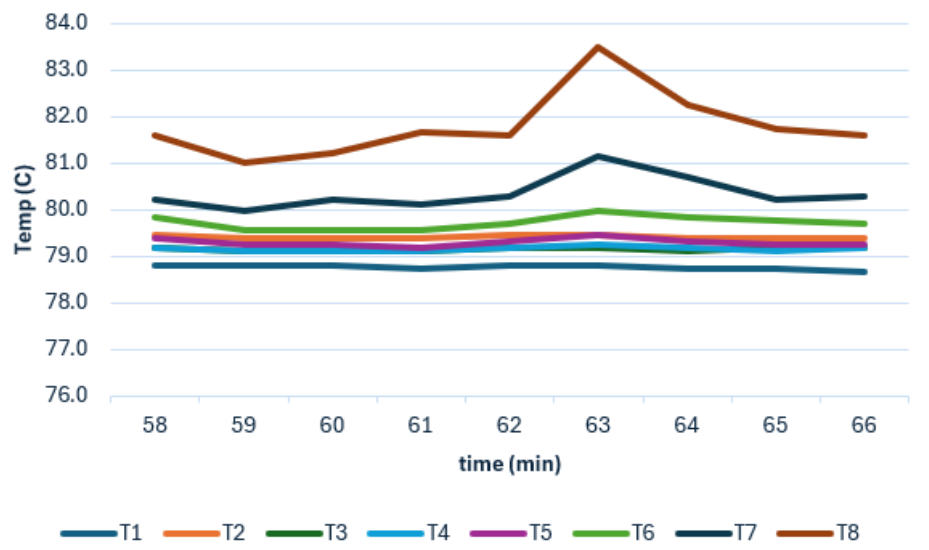


Table 2 summarizes the measured boil-up rates, pressure drops, and degrees of foaming. The reported values are the average of three trials, where the trials were compared for error analysis. Increasing reboiler duty resulted in higher vapor generation rates and greater pressure drops across the column, thus showing a proportional relationship with between reboiler duty, pressure drop, and boil-up rate. The foaming intensity followed the same trend; ranging from gentle bubbling at 0.75 kW to localized foaming at 1.0 kW and global violent foaming at 1.25 kW. The sharp jump in foaming between 1.0 kW and 1.25 kW suggests that running the column above 1.25 kW would greatly lower efficiency.

Table 2: Experiment A result summary

Reboiler Duty (kW)	Pressure drop (cm H ₂ O)	Log Pressure Drop	Boil-up Rate (L/hr)	Log Boil-up Rate	Degree of Foaming
1.25	115	2.06	3.75	0.574	Gentle
1.0	90	1.95	2.98	0.474	Localized violent
0.75	75	1.88	1.88	0.272	Global violent

Figure 7 presents the log ΔP vs. log boil-up rate relationship. The data exhibit a near-linear correlation, confirming that the pressure drop increases proportionally with vapor velocity as predicted by Bernoulli's principle. As vapor flow strengthens, both kinetic and frictional losses through the column rise, leading to the observed increase in ΔP . At the highest duty, foaming and entrainment contribute additional hydrodynamic resistance, signaling the onset of flooding behavior.

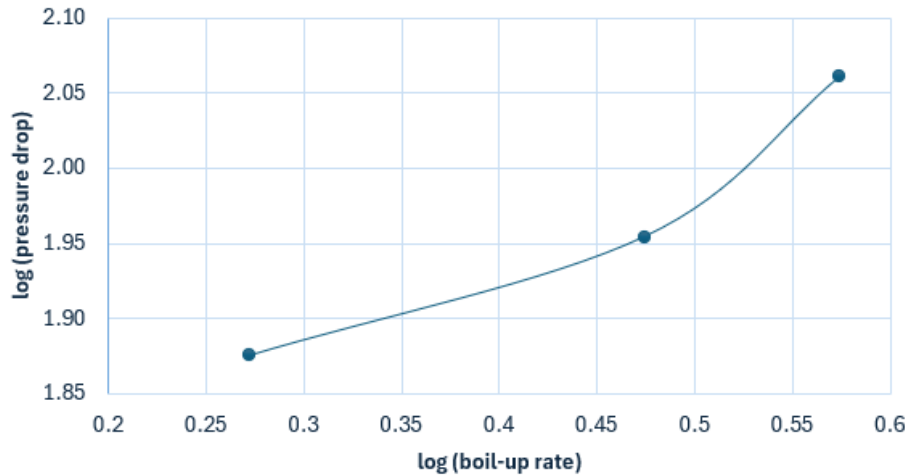
Figure 7: $\log \Delta P$ vs $\log$ boil-up rate relationship

Table 3 summarizes the vapor–liquid equilibrium (VLE) data for the distillate and bottoms at each reboiler power setting.

Here, Y_{EtOH} and X_{EtOH} represent the ethanol mole fractions in the vapor and liquid phases, respectively, while Y_{H_2O} and X_{H_2O} represent those for water. K_A and K_B are the phase equilibrium ratios ($K = Y/X$) for ethanol and water, and α_{AB} is the relative volatility between the two components, defined as $\alpha_{AB} = K_A/K_B$.

Table 3: VLE data for bottoms and distillate

Duty (kW)	location	Y_{EtOH}	X_{EtOH}	Y_{H_2O}	X_{H_2O}	K_A	K_B	α_{AB}
1.25	distillate	0.811	0.809	0.189	0.191	1.002	0.990	1.01
	bottoms	0.325	0.042	0.675	0.958	7.796	0.705	11.06
1.0	distillate	0.797	0.782	0.203	0.218	1.020	0.929	1.10
	bottoms	0.342	0.044	0.658	0.956	7.806	0.688	11.35
0.75	distillate	0.787	0.761	0.213	0.239	1.034	0.892	1.16
	bottoms	0.420	0.054	0.580	0.946	7.841	0.612	12.80

The data show that as power increases, the mole fraction of ethanol in the bottoms ($X_{EtOH,b}$) decreases slightly, while the distillate composition ($Y_{EtOH,d}$) increases marginally. This means that higher vapor flow improves separation in the upper trays, as more ethanol is stripped upward. The relative volatility (α_{AB}) between ethanol and water decreases slightly with increasing power, suggesting the column approaches equilibrium more quickly at higher boil-up rates, leaving less compositional difference between phases.

Table 4 summarizes the average relative volatility, theoretical number of plates (N), and column efficiency for each reboiler duty. Here, $N + 1$ represents the total number of theoretical stages, including the reboiler, and N is the number of trays above the reboiler. Column efficiency was calculated from the ratio of actual to theoretical plates.

Table 4: Average volatility, number of plates, and efficiency

Duty (kW)	$\alpha_{AB,avg}$	$N + 1$	N	Efficiency (%)
1.25	3.35	3.79	2.79	31.0
1.0	3.53	3.46	2.46	27.3
0.75	3.85	2.99	1.99	22.1

As can be seen on the table, as reboiler duty increases, the average volatility decreases slightly, while both the number of effective plates and efficiency increase. This indicates that higher power improves vapor–liquid contact and mass transfer between trays. However, the reduction in volatility reflects that the system becomes closer to equilibrium at higher vapor rates, leading to smaller composition differences per stage.

Figure 8: Efficiency vs. Boil-up Rate

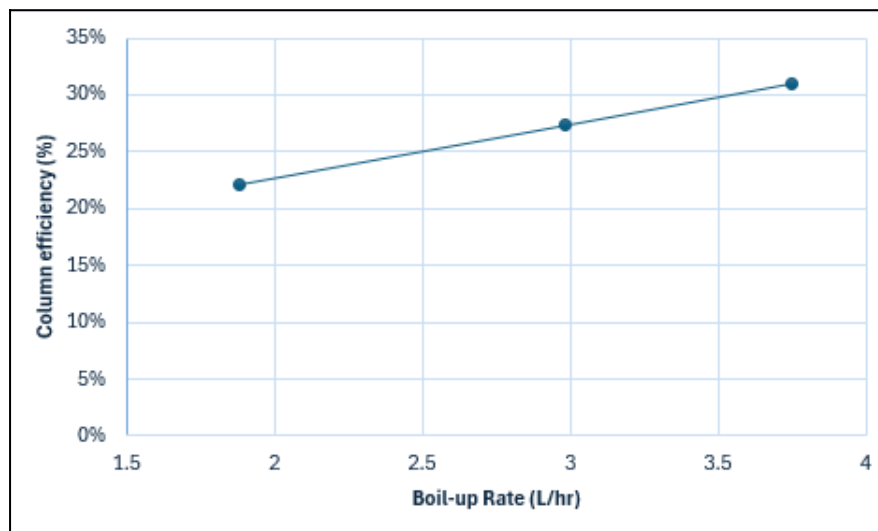


Figure 8 (Efficiency vs. Boil-up Rate) shows this relationship graphically. Efficiency increases steadily with boil-up rate, but the slope begins to taper at higher rates. If extrapolated, the curve suggests that efficiency would eventually plateau or decrease beyond 1.25 kW due to excessive foaming and flooding, which interfere with separation.

Overall, the data from Experiment A show that increasing reboiler power enhances separation up to a point, but excessive vapor flow would likely reduce efficiency due to hydrodynamic instability

Experiment B

This part of the experiment investigated how changing the reflux ratio affects the composition of the top and bottom products and the overall column efficiency during batch distillation. The column was operated at a constant reboiler duty of 0.75 kW, and two reflux ratios (5:1 and 12:1) were tested after steady-state conditions were reached. Increasing the reflux ratio returns more condensed liquid to the column, which enhances separation efficiency but requires more time and energy. The same calculations and methods used in Experiment A (for determining relative volatility, number of plates, and efficiency) were repeated here, but only the essential averaged results are reported below.

Table 5: Moles of ethanol and water in bottoms and distillate

Reflux Ratio	location	Composition (%v/v)	Moles EtOH	Moles H ₂ O	Mole Fraction EtOH
5:1	Distillate	90.27	1.546	0.540	0.7441
	Bottoms	12.94	0.222	4.834	0.0438
12:1	Distillate	92.77	1.589	0.402	0.799
	Bottoms	13.56	0.232	4.80	0.046

As shown in Table 5, increasing the reflux ratio from 5:1 to 12:1 slightly raised the ethanol mole fraction in the distillate (from 0.744 to 0.799), while the bottom composition remained nearly unchanged. This indicates that a higher reflux ratio improves separation by allowing better contact between the vapor and liquid phases, resulting in a purer top product. However, the improvement becomes smaller at higher ratios, meaning that increasing reflux beyond this point would offer diminishing returns while consuming more energy and time. The results of the efficiency calculations, performed using the same method as in Experiment A, are summarized in Table 6.

Table 6: Column efficiency when varying reflux ratio

Reflux Ratio	$\alpha_{AB,avg}$	$N+1$	N	Efficiency (%)
5:1	3.691	3.187	2.187	24.1
12:1	3.479	3.545	2.545	28.0

As seen in Table 7, the column efficiency increases from 24.1 % to 28.0 % as the reflux ratio rises from 5:1 to 12:1. This trend shows that a higher reflux ratio improves the vapor–liquid contact within the column, enhancing mass transfer and producing a purer distillate. However, the improvement is modest, suggesting that the system approaches its practical efficiency limit near a 12:1 reflux ratio. Beyond this point, further increases would yield only small efficiency gains while significantly raising operating time and energy consumption.

Overall, the results from Experiments A and B show that both increasing the reboiler duty and reflux ratio enhance the separation efficiency of the ethanol–water system, though with diminishing improvements beyond moderate operating conditions due to practical energy and time limitations.

5. Discussion (15 pts)

Experiment A began at a wattage of 1.25kW being transmitted directly into the reboiler. Steady state was achieved by monitoring the system for 20+minutes and watching the temperature profile begin to stagnate. From there, the electricity was decreased in 0.25kW increments to 1.00kW and 0.75kW. During trials, samples were taken from the reboiler to analyze the mixture's composition to determine how effective the settings were. Three sets of data was collected for each setting. Ethanol concentrations inside the samples decreased over time, which was expected, as it has the lowest boiling point. Additionally, time collecting distillate was recorded during sampling, enabling calculations for the boil-up rate. In this experiment, a positive correlation was found between the power provided to the reboiler and the boil-up rate, implying that increasing the power of the reboiler directly increases the flowrate of distillate generated. This can be seen in *Table 2*. An increase in pressure drop can also be seen with the increase of boil-up rate due to the increase of power to the reboiler. *Figure 7* shows how the log pressure differential across the column increases almost linearly with the log boil-up rate, which is as anticipated. As the boil-up rate goes up, the vapor pressure and velocity increase according to Bernoulli's Principle. With the increase in velocity, the vapor gets more violent, making the flow significantly more turbulent and chaotic. This led to much more foaming in the column. Another relationship found in Experiment A was that the column efficiency grew with increased power to the reboiler. *Figure 8* showcases how larger boil-up rates (caused by increase reboiler power) lead to increased column efficiency from 22.1% at 0.75kW of power to 31% at 1.25kW. The slope of this relationship began to taper at higher boil-up rates, suggesting that there are diminishing returns on column efficiency when one increases the reboiler duty. From findings on pressure drop and efficiency, the experiment proves that there is a point at which increasing the power to the reboiler produces too high of a pressure drop, leading to diminishing returns on column efficiency. This indicates that there is an ideal point at which the column's reboiler can be optimized.

Experiment B stayed at a constant wattage of 0.75kW while the reboiler rate was varied at a 5:1 ratio and 12:1 ratio. Again, three sets of data were collected at each setting. The key hypothesis tested in this experiment was that a higher reflux ratio would lead to a more efficient separation of ethanol from water. Like Experiment A, samples were taken from the bottom and top products to measure the concentration of ethanol after the system reached steady state. From the data collected, changing the reflux ratio from 5:1 to 12:1 did not make a significant difference in the effectiveness of the column. The number of stages barely increased and the efficiency of the column went from 24.1% to 28%. While this proved that a higher reflux ratio leads to a more efficient separation, it did not do so as drastically as expected.

In theory, increasing the reflux ratio by this amount should drastically increase the efficiency of the column while sacrificing energy and time. The McCabe-Thiele Method for distillation columns highlights how the concentration of bottom and top product can be found through a simple, graphical method [3]. Here, the vapor composition of methanol can be found from the following equation(s):

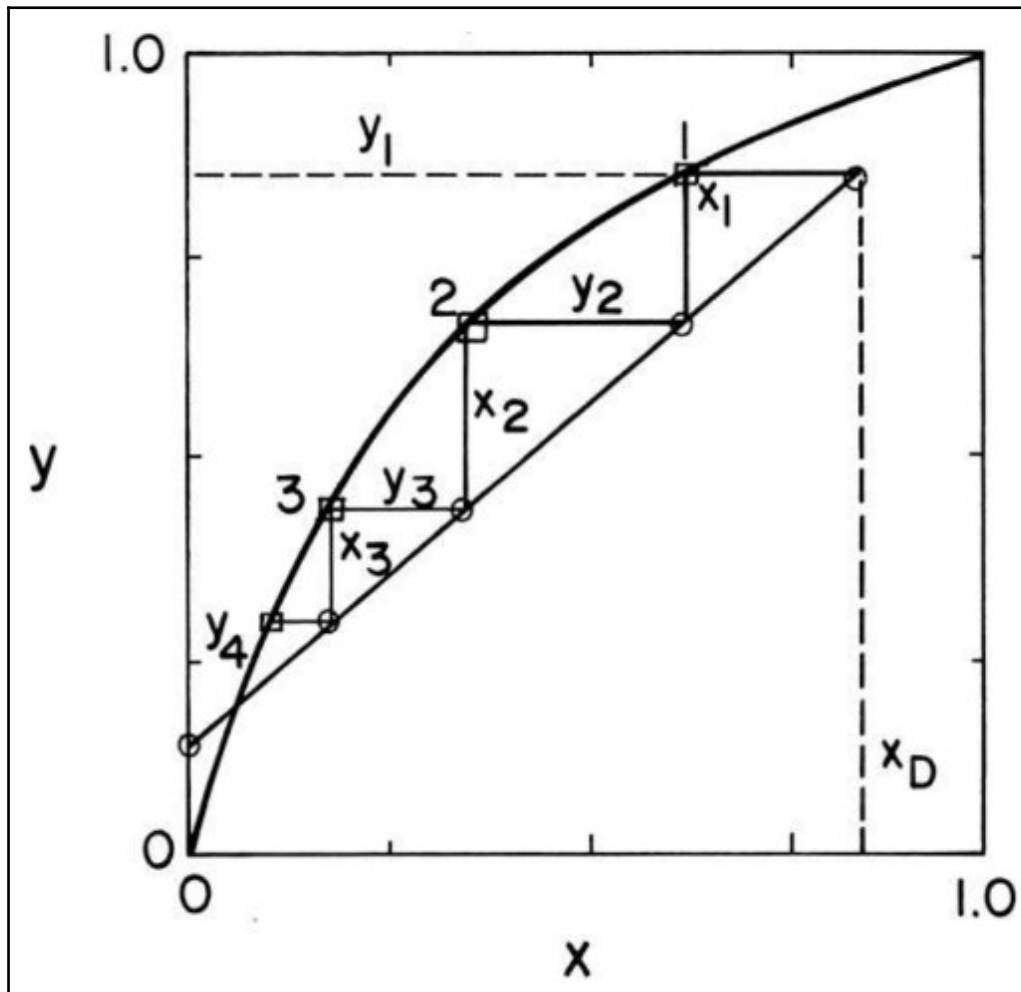
Figure 9: McCabe Thiele Top Operating Equation

$$y = \frac{L}{V}x + \left(1 - \frac{L}{V}\right)x_D \quad (4-23)$$

$$\frac{L}{V} = \frac{L}{L+D} = \frac{L/D}{L/D+1} \quad (4-26)$$

In the McCabe-Thiele Method, the vapor composition can be found by graphing the equilibrium data and “operating lines” and finding where they intersect. Additionally, the number of stages can be determined by connecting the operating line and equilibrium line together in a concerted fashion shown in Figure 10:

Figure 10: Stage Calculation Using McCabe Thiele



From the equations and graph, it is clear to see that increasing the reflux ratio (L/D) results in a decrease in the slope of the operating line. If the slope of the operating line decreases, the number of stages found graphically will also decrease by a large margin.

This McCabe-Thiele theory validates that increasing the reflux ratio will result in a higher column efficiency.

The data found in Experiment B strays away from the McCabe-Thiele theory for distillation, as a large change in reflux ratio did not result in a large change in number of stages/efficiency. This can be due to multiple reasons. Firstly, the McCabe-Thiele theory assumes constant molar overflow, perfect vapor-liquid equilibrium in every tray, no heat loss, constant relative volatility, and perfectly mixed trays. In this experiment, there is no true way to ensure these ideal conditions are met. Heat is lost to the column and surrounding air, trays receive differing amounts of vapor/liquid throughout the column, and volatility changes due to pressure drop across the column. The reduced efficiency observed experimentally in the distillation column is a direct consequence of non-ideal conditions, which can be validated by conducting an energy balance on the system. Under steady-state conditions, simplifying the balance by assuming the feed and product enthalpies cancel reveals that the energy lost from the system equals the difference between the reboiler and condenser enthalpies. Since the condenser removed less heat than the reboiler supplied (calculated from their respective duties and mass flow rates), a significant amount of energy was lost. These thermal losses reduced the heat available to drive vapor-liquid equilibrium on each tray, thus decreasing the overall column efficiency. Consequently, the actual separation was less effective than predicted by ideal theoretical models like McCabe-Thiele.

Energy loss at this level could be the result of this experiment being taken at lab-grade, which is cheaper and takes on a much smaller production of product. At industrial-grade, a lot more work is put into making the column well insulated and to ensure all the equipment, including the trays, are as efficient as possible. Thus, an industrial-grade distillation column may result in data that is much closer to theoretical values for column efficiency.

6. References (10 pts)

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- [6] Armfield Limited, UOP3CC Continuous Distillation Column Instruction Manual, Issue 38 March 2014.
- [7] Stichlmair, J. G., Klein, H., & Rehfeldt, S. (2021). *Distillation Principles and Practice*. Newark American Institute Of Chemical Engineers Ann Arbor, Michigan Proquest.

7. Appendix I: Data Tabulation/Graphs (10 pts)

The following section presents the experimental data collected during both Experiments A and B. For each reboiler duty and reflux ratio, the pressure, distillate volume, collection time, ethanol concentration (% v/v) in the top and bottom products, and the corresponding boil-up rate were recorded. While all data is reported here, the averages are used for calculations, while the top and bottom ethanol mole fractions for each sample are compared in the error analysis. The temperatures across the column as a function of time are omitted due to the excessive amount of data and the fact that these approximate temperatures can be observed in Figure 3-6.

Table 7: Experiment A raw data

Duty (kW)	Sample	Pressure drop (cm H ₂ O)	Volume (mL)	Time (s)	Distillate EtOH conc. (%v/v)	Bottoms EtOH conc. (%v/v)	Boil-up Rate (L/hr)
1.25	1	113	8.4	7.96	93.37	12.67	3.82
	2	117	7.9	7.72	93.04	12.03	3.69
	3	115	8.2	7.81	93.19	12.32	3.73
	Avg	115	8.2	7.84	93.21	12.35	3.75
1.0	1	93	6.1	7.15	92.02	14.31	3.07
	2	87	6.2	7.72	92.11	11.54	2.89
	3	90	6.15	7.49	92.08	12.62	2.94
	Avg	90	6.15	7.44	92.07	12.93	2.98
0.75	1	72	8.9	17.9	91.23	16.11	1.79
	2	78	8.7	15.9	91.09	14.92	1.97
	3	74	8.8	16.8	91.14	15.39	1.89
	Avg	75	8.8	16.9	91.16	15.52	1.88

Table 8: Experiment B raw data

Reflux Ratio	Sample	Pressure drop (cm H ₂ O)	Volume (mL)	Time (s)	Distillate EtOH conc. (%v/v)	Bottoms EtOH conc. (%v/v)	Boil-up Rate (L/hr)
5:1	1	85	5.8	15.50	93.04	13.60	1.347
	2	83	6.9	17.15	87.49	12.27	1.448
	3	84	6.4	16.48	90.83	12.31	1.387
	Avg	84	6.35	16.33	90.27	12.35	1.398
12:1	1	86	6	11.88	93.61	14.38	1.818
	2	85	5.5	12.03	91.92	12.74	1.646
	3	85	6	11.92	92.61	13.39	1.712
	Avg	85.5	5.75	11.96	92.77	13.56	1.732

8. Appendix II: Error Analysis (10 pts)

Qualitative error analysis

There are several sources of error that may have affected the measured compositions and column performance. One source of error is uncertainty in preparing the initial 10wt% ethanol water feed mixture. Small changes in the concentration would affect both the distillate and bottom compositions, on top of that if the column temperatures were not stabilized before sampling, the measured compositions may not reflect true equilibrium values, since ethanol mole fraction is dependent on the temperature.

How the group took the samples could also have affected our results. If the return line did not stay filled with liquid, vapor could enter the line and mix with the sample, which would change the composition we measured. Even when sampling is done correctly, small timing differences can lead to variation, and that is because distillate composition can change quickly during batch distillation. At higher boil up rates, the behavior inside the column may become less steady, and increasing vapor flows can lead to foaming or uneven liquid distribution on the trays, which affects how well the vapor and liquid mix. This can cause changes in the pressure drop, and separation performance.

Overall, these factors can create small differences between measurements, but they won't change the trends observed in the experiment. The relationships between boil up rate, product separation, and reflux ratio were still clear even with some variations.

Table 1. Standard deviation results for different power levels.

Power (kW)	Average Top Ethanol Mol Frac $\pm$ Stdev	Average Bottom Ethanol Mol Frac $\pm$ Stdev
0.75	0.761 $\pm$ 0.00224	0.0536 $\pm$ 0.00326
1	0.782 $\pm$ 0.001487	0.0438 $\pm$ 0.00729
1.25	0.809 $\pm$ 0.00570	0.0416 $\pm$ 0.00169

Table 2. Standard deviation results for different reflux ratios.

Reflux Ratio	Average Top Ethanol Mol Frac $\pm$ Stdev	Average Bottom Ethanol Mol Frac $\pm$ Stdev
5:1	0.744 $\pm$ 0.0860	0.0438 $\pm$ 0.00350
12:1	0.798 $\pm$ 0.0287	0.0462 $\pm$ 0.00436

9. Appendix III: Sample Calculations (10 pts)

All calculations below are shown for the 1.25 kW reboiler duty in Experiment A. The same steps were repeated for the other reboiler duties (0.75 and 1.0 kW) and for both reflux ratios in Experiment B; only the numerical values change.

- **Preparing 10 wt% ethanol solution**

To begin, the total volume required for a 10 wt% ethanol solution was determined. For this calculation, the densities of ethanol and water are:

$$\rho_{\text{EtOH}} = 0.789 \frac{\text{kg}}{\text{L}}, \rho_{\text{H}_2\text{O}} = 1.0 \frac{\text{kg}}{\text{L}}$$

Assuming a 1 kg basis:

$$0.1 \text{ kg EtOH} \times \frac{1 \text{ L}}{0.789 \text{ kg}} = 0.127 \text{ L EtOH} \quad 0.9 \text{ kg H}_2\text{O} \times \frac{1 \text{ L}}{1.0 \text{ kg}} = 0.9 \text{ L H}_2\text{O}$$

Total mixture volume:

$$0.127 \text{ L EtOH} + 0.900 \text{ L H}_2\text{O} = 1.027 \text{ L total}$$

The volume percent ethanol is therefore:

$$\frac{0.127}{1.027} \times 100 = 12.3 \%$$

This shows that to prepare 10 L of a 10 wt% ethanol solution, 1.23 L of ethanol and 8.77 L of water are required.

- **Boil-up rate at 1.25 kW**

From Table 7 (1.25 kW row, averaged over two samples):

- Average collected volume of distillate: $V = 8.2 \text{ mL}$
- Average collection time: $t = 7.84 \text{ s}$

First convert to a volumetric flow rate in L/hr:

$$\dot{V} = \frac{8.2 \text{ mL}}{7.84 \text{ s}} \times \frac{1 \text{ L}}{1000 \text{ mL}} \times \frac{3600 \text{ s}}{1 \text{ hr}} \quad \dot{V} = 3.75 \text{ L/hr}$$

This is the boil-up rate reported for the 1.25 kW case.

- **Converting distillate and bottoms compositions to mole fractions**

Average distillate and bottoms ethanol concentrations at 1.25 kW from Table 7:

- Distillate: 93.21 %EtOH (v/v)
- Bottoms: 12.35 %EtOH (v/v)

Take a 100 mL basis for each stream.

Distillate (top product):

Volumes:

$$V_{\text{EtOH,d}} = 93.21 \text{ mL}, V_{\text{H}_2\text{O,d}} = 100 - 93.21 = 6.79 \text{ mL}$$

Masses (using densities above):

$$m_{\text{EtOH},d} = 93.21 \times 0.789 = 73.5 \text{ g} \quad m_{\text{H}_2\text{O},d} = 6.79 \times 1.00 = 6.79 \text{ g}$$

Moles (using $M_{\text{EtOH}} = 46.07 \text{ g/mol}$,
 $M_{\text{H}_2\text{O}} = 18.01 \text{ g/mol}$):

$$n_{\text{EtOH},d} = \frac{73.5}{46.07} = 1.60 \text{ mol} \quad n_{\text{H}_2\text{O},d} = \frac{6.79}{18.01} = 0.38 \text{ mol}$$

Total moles:

$$n_{\text{tot},d} = 1.60 + 0.38 = 1.98 \text{ mol}$$

Liquid mole fractions in the distillate:

$$x_{\text{EtOH},d} = \frac{1.60}{1.98} = 0.809, \quad x_{\text{H}_2\text{O},d} = 1 - x_{\text{EtOH},d} = 0.191$$

Bottoms (bottom product):

Volumes:

$$V_{\text{EtOH},b} = 12.35 \text{ mL}, \quad V_{\text{H}_2\text{O},b} = 87.65 \text{ mL}$$

Masses:

$$m_{\text{EtOH},b} = 12.35 \times 0.789 = 9.74 \text{ g} \quad m_{\text{H}_2\text{O},b} = 87.65 \times 1.00 = 87.65 \text{ g}$$

Moles:

$$n_{\text{EtOH},b} = \frac{9.74}{46.07} = 0.21 \text{ mol} \quad n_{\text{H}_2\text{O},b} = \frac{87.65}{18.01} = 4.87 \text{ mol}$$

Total:

$$n_{\text{tot},b} = 0.21 + 4.87 = 5.08 \text{ mol}$$

Liquid mole fractions in the bottoms:

$$x_{\text{EtOH},b} = \frac{0.21}{5.08} = 0.04, \quad x_{\text{H}_2\text{O},b} = 1 - x_{\text{EtOH},b} = 0.96$$

These mole fractions for distillate and bottoms are the values used in the volatility and plate-number calculations.

- **Using VLE data to obtain vapor compositions and distribution coefficients**

The vapor–liquid equilibrium (VLE) data for the ethanol–water system were obtained from Source 7 (ACS J. Chem. Eng. Data, 2020) and used to estimate vapor-phase compositions at the liquid-phase mole fractions measured during the experiment. Because our experimental liquid mole fractions (x_{EtOH}) did not always match the tabulated values exactly, linear interpolation between adjacent data points was used to find the corresponding vapor-phase mole fraction (y_{EtOH}).

At $x_{\text{EtOH},d} = 0.809$, $y_{\text{EtOH},d} = 0.811$ (from VLE data interpolation).

Because only ethanol and water are present,

$$x_{\text{H}_2\text{O},d} = 1 - x_{\text{EtOH},d} = 0.191, \quad y_{\text{H}_2\text{O},d} = 1 - y_{\text{EtOH},d} = 0.189.$$

The distribution coefficients (K-values) for the distillate are then

$$K_{A,d} = \frac{y_{\text{EtOH},d}}{x_{\text{EtOH},d}} = \frac{0.811}{0.809} = 1.002, \quad K_{B,d} = \frac{y_{\text{H}_2\text{O},d}}{x_{\text{H}_2\text{O},d}} = \frac{0.189}{0.191} = 0.99.$$

The same procedure is applied to the bottoms composition to obtain the bottoms compositions, $K_{A,B}$, and $K_{B,B}$. From the K-values, the relative volatility of ethanol to water is

$$\alpha_{AB,d} = \frac{K_{A,d}}{K_{B,d}}, \alpha_{AB,b} = \frac{K_{A,b}}{K_{B,b}}.$$

Where component A is ethanol and component B is water, and “d” denotes distillate, while “b” denotes bottoms. For the representative case, $\alpha_{AB,D} = 1.012$ and $\alpha_{AB,B} = 11.064$, giving an average relative volatility,

$$\alpha_{AB,avg} = \sqrt{\alpha_{AB,d} \alpha_{AB,b}} = \sqrt{1.012 \times 11.064} = 3.346.$$

- **Number of theoretical plates and column efficiency**

The number of theoretical plates N is obtained from the Fenske relation for a binary system at total reflux:

$$N + 1 = \frac{\log \left[\left(\frac{x_{EtOH}}{x_{H_2O}} \right)_d \left(\frac{x_{H_2O}}{x_{EtOH}} \right)_b \right]}{\log (\alpha_{AB,avg})}.$$

Using the average distillate and bottoms compositions

($x_{EtOH,d} = 0.809$, $x_{H_2O,d} = 0.191$, $x_{EtOH,b} = 0.042$, $x_{H_2O,b} = 0.958$) and $\alpha_{AB,avg} = 3.35$,

$$N + 1 = \frac{\log \left[\left(\frac{0.809}{0.191} \right) \left(\frac{0.958}{0.042} \right) \right]}{\log (3.35)} = 3.79,$$

So,

$$N = 3.79 - 1 = 2.79 \text{ theoretical plates.}$$

The column has 9 physical trays, so the column efficiency for this run is:

$$\text{Efficiency} = \frac{N}{N_{\text{actual}}} \times 100 = \frac{2.79}{9} \times 100 = 31 \%.$$

The same sequence of steps was repeated for the other reboiler powers in Experiment A and for both reflux ratios in Experiment B.

10. Appendix IV: Individual Team Contributions

	Time (Hours)			
	Joshua Olusoji	Omar Zaruk	Sanad Abu Hweij	Qussay Suleiman
Operator (Both Lab Days)	2	4	2	3
Pre-Lab Editing	1	1	2	2
Experimental Objective (Prelab)				1
Apparatus (prelab)	2			

Materials & Supplies (prelab)			1	
Procedure (prelab)		3		
Project Safety Evaluation (prelab)				2
Final Lab Editing	1	1		1
Executive Summary (final lab)		1	2.5	
Introduction (Final lab)				
Theory (Final Lab)		4		
Results (final lab)		1		3
Discussion (final lab)	4			
References (final and/or prelab)	1			0.5
Data Tabulation/Graphs (final)				1
Error Analysis (final lab)			3	
Sample Calculations (final Lab)				2
Power Point Presentation	2		2	2
Total	13	14	12.5	17.5